Thesis Progress Report

What I've been working on since Week 9

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Contents

1 Overview **2**

2 The R-Squared Puzzle **3**

3 Strict Exogeneity Test **4**

4 Why I Didn't Use First Differencing **5**

5 Cointegration — I Have a Question Here **6**
5.1 What I found . . . 6
5.2 My question . . . 6

6 Panel Unit Root Tests **7**
6.1 Levels . . . 7
6.2 First Differences . . . 7

7 GMM Evaluation **8**
7.1 What I was checking for . . . 8
7.2 Results . . . 8
7.3 Why I think GMM is still unreliable here . . . 8

8 Results Summary **9**
8.1 TWFE Progression . . . 9
8.2 Preferred Specification . . . 9
8.3 Robustness . . . 9

9 Questions for Our Meeting **10**

1 Overview

This is a summary of what I havve done since our Week 9 meeting (18 March). I have tried to be honest about what worked, what didn't, and what I'm still unsure about.

The main things I tackled:

1. **The R-squared puzzle** — why TWFE gives lower R^2 than one-way FE
2. **Strict exogeneity test** — formally tested whether the static within estimator is consistent
3. **Why first differencing doesn't help** — connecting it to the exogeneity test result
4. **Cointegration** — I'm still not fully sure what to do here and have a question for you
5. **Panel unit root tests** — IPS, LLC, and Fisher-ADF on all four variables
6. **Systematic GMM evaluation** — five different lag structures, all documented

2 The R-Squared Puzzle

I noticed that the two-way FE model has a *lower* within-R^2 (0.203) than the one-way FE model (0.215), which seems wrong since TWFE nests the one-way specification. I looked into this and it turns out it's not a bug — it's just how `plm` calculates within-R^2.

The within-R^2 formula applied by `plm` is:

$$R^2_{\text{within}} = 1 - \frac{\text{RSS}_{\text{demeaned}}}{\text{TSS}_{\text{demeaned}}}$$

, but both RSS and TSS are computed on the **demeaned** data, and the demeaning is different between the two models. When we add year fixed effects, the TSS shrinks (because dummies absorb national-level variation from the total), but the RSS shrinks by slightly less. So the ratio gets worse even though the fit is actually better in absolute terms.

Here's the numbers:

Metric	One-Way FE (state only)	Two-Way FE (state + year)
TSS (demeaned)	4.170	3.589 (−14%)
RSS (demeaned)	3.275	2.858 (−13%)
Within R^2	**0.215**	**0.203**

TSS fell by 14% when year dummies absorbed national-level shocks, but RSS only fell by 13%. Because the denominator shrank faster than the numerator, R^2 mechanically dropped. The model does fit better in absolute terms — RSS is lower. The within-R^2 values just aren't comparable across models with different demeaning.

Conclusion: No code correction needed. The two-way FE is still the right specification.

Question: Do we need to add this part to the thesis?

price at any point in time is uncorrelated with the error at any other point in time. Past, present, and future.

3 Strict Exogeneity Test

I hadn't formally tested this before, so I ran a leads test on the static TWFE.

The idea is: if strict exogeneity holds, then one-period-ahead values of the regressors should be insignificant when added to the model. **If they're significant**, it means current errors are correlated with future regressors — i.e., there's feedback from demand shocks to future prices.

The augmented model I estimated:

$$\ln Q_{it} = \beta_1 \ln P^e_{it} + \beta_2 \ln P^{ng}_{it} + \beta_3 \ln \mathrm{GDP}_{it} + \underbrace{\gamma_1 \ln P^e_{i,t+1} + \gamma_2 \ln P^{ng}_{i,t+1} + \gamma_3 \ln \mathrm{GDP}_{i,t+1}}_{\text{leads}} + \alpha_i + \lambda_t + u_{it}$$

Then I tested $H_0 : \gamma_1 = \gamma_2 = \gamma_3 = 0$ jointly (HC1 SEs clustered by state):

χ^2 statistic	df	p-value
17.70	3	**0.0005**

Rejected decisively. So **strict exogeneity fails** — demand shocks do feed back into next year's prices, which is consistent with what you'd expect (a demand spike one year leads utilities to adjust prices the next). This makes the static within estimator technically inconsistent here.

Why I think it Matters: it rules out first differencing, which I explain in the next section.

So far:

Static WE: Strict exogeneity violated
Dynamic within (FE + lag): Same as above, plus Nickell bias from the lagged DV

—> add to motivation of GMM

4 Why I Didn't Use First Differencing

First differencing is sometimes proposed as a simpler fix for non-stationarity issues. But FD requires **sequential** exogeneity:

$$\mathbb{E}[\Delta x_{it} \cdot \Delta v_{it}] = 0$$

which is equivalent to saying current regressors are uncorrelated with *last period's* error. The strict exogeneity test above confirms that $x_{i,t+1}$ is correlated with v_{it} — which means x_{it} is correlated with $v_{i,t-1}$. So the FD assumption also fails.

The idea is: if there's feedback from demand to prices, FD can't fix it. We need GMM with deep lags, which instruments with $\ln Q_{i,t-2}$ and further back — a much weaker condition that the feedback shouldn't violate.

5 Cointegration — I Have a Question Here

5.1 What I found

Since the unit root tests (next section) showed that $\ln P^e$ and $\ln \mathrm{GDP}$ are I(1) while $\ln Q$ and $\ln P^{ng}$ are I(0), I needed to check whether the levels regression is valid or spurious.

I ran a residual-based cointegration test: extract residuals from the static TWFE, then test whether those residuals are stationary.

run a level regression

Test	IPS on TWFE residuals	Fisher-ADF on TWFE residuals
Statistic	NA	391.047
p-value	NA	$< \mathbf{0.001}$
Decision	(failed to compute)	**COINTEGRATED**

The Fisher test **strongly rejects** the null that residuals have a unit root. If residuals are I(0), the variables share a long-run equilibrium and the levels regression is valid.

5.2 My question

it assumed all variables are I(1) instead of mix

My concern: the *Engle-Granger* residual approach is technically inappropriate when integration orders are *mixed* (some I(0), some I(1)). So I am wondering, if *Johansen cointegration* or an *ARDL bounds test* would be a better choice instead. I'm not sure how to handle this.

Is the Fisher residual test result sufficient here, or do I need to implement Johansen / ARDL bounds? I'm happy to do either but I wanted to check before going down that path, since it's territory that wasn't covered in the MPhil courses and I'd need to study it.

recall: previous GMM output, ρ = 1.029, which looked like an explosive process —> decided to do an unit root test

6 Panel Unit Root Tests

I ran IPS, LLC, and Fisher-ADF on all four log variables, in levels and first differences. Lag lengths selected by AIC ($p_{\max} = 3$ for levels, $p_{\max} = 2$ for first differences).

6.1 Levels

Variable	IPS W-stat (p)	LLC t^* (p)	Fisher χ^2 (p)	Classification
$\ln(Q)$	-2.464 (0.007)	-1.272 (0.102)	268.455 (0.000)	**I(0)**
$\ln(P^e)$	4.718 (1.000)	2.999 (0.999)	136.952 (0.012)	**I(1)**
$\ln(P^{ng})$	-5.613 (0.000)	-7.282 (0.000)	361.272 (0.000)	**I(0)**
ln(GDP)	11.456 (1.000)	6.117 (1.000)	105.577 (0.384)	**I(1)**

I(0) if at least 2 of 3 tests reject H_0 at 5%.

$\ln(Q)$ and $\ln(P^{ng})$ are stationary. $\ln(P^e)$ and ln(GDP) are I(1) — this is what triggered the need for the cointegration check above.

6.2 First Differences

All four first-differenced series are strongly stationary:

Variable	IPS (p)	Fisher (p)
$\Delta \ln(Q)$	0.000	0.000
$\Delta \ln(P^e)$	0.000	0.000
$\Delta \ln(P^{ng})$	0.000	0.000
$\Delta \ln(\text{GDP})$	0.000	0.000

This confirms $\ln P^e$ and $\ln \text{GDP}$ are I(1) rather than I(2) or higher.

One thing the unit root results do clarify: the $\hat{\rho} > 1$ I got in the GMM output earlier was finite-sample bias, not a genuine unit root in the data. $\ln Q$ is stationary in levels.

7 GMM Evaluation

You asked me to experiment with different lag structures. I tested 5 Arellano-Bond specifications and documented the diagnostics for each.

7.1 What I was checking for

A specification passes if:

- $\hat{\rho} < 1$ (demand process isn't explosive)
- Sargan $p \in (0.05,\ 0.95)$ (instruments are valid but not over-proliferated)
- AR(2) $p > 0.10$ (no second-order serial correlation in the differenced residuals)

7.2 Results

Specification	rho	beta	SE	Sargan	AR2	Verdict
Diff GMM, lags 2:3	0.279	-0.102	0.023	0.160	0.170	Pass
Diff GMM, lags 2:4	0.632	-0.113	0.017	0.319	0.723	Pass
Diff GMM, lags 2:5	0.675	-0.060	0.035	0.974	0.769	Fail
Diff GMM, lags 3:5	0.641	-0.078	0.025	0.528	0.814	Pass
Sys GMM, lags 2:3	0.990	-0.037	0.005	0.150	0.744	Pass

Sys GMM passes the formal tests but $\hat{\rho} = 0.990$ is implausibly close to a unit root — this feels like residual instrument proliferation rather than a genuine result.

7.3 Why I think GMM is still unreliable here

The $\hat{\rho}$ estimates jump around a lot:

- 0.279 for lags 2:3
- then 0.632 for lags 2:4
- then 0.641 for lags 3:5

If GMM were identified, I'd expect more stability across lag choices. The instability tells me the instrument count is still too close to $N = 51$ even with tight lag windows.

The general rule is that instruments should be well below N. Even lags 2:3 pushes close to that limit once year dummies are included. This is a fundamental limitation of having only 51 states — it's not something I can fix by choosing different lags.

So my conclusion (from analysis) is the same as before: **it seems like, the within estimator is the best feasible option at this sample size**. What's different now is that I've tested this systematically across 5 specifications rather than just reporting the one that happened to converge first.

8 Results Summary

For completeness, here are the preferred estimates.

8.1 TWFE Progression

Model	TSS	RSS	R^2
Static, one-way FE	4.170	3.275	0.215
Static, two-way FE	3.589	2.858	0.203
Dynamic, one-way FE	3.536	1.070	0.697
Dynamic, two-way FE	**3.003**	**0.799**	**0.734**

8.2 Preferred Specification

$$\ln Q_{it} = \underbrace{0.824^{***}}_{(0.061)} \ln Q_{i,t-1} \underbrace{- 0.195^{**}}_{(0.067)} \ln P^e_{it} + \underbrace{0.048^{*}}_{(0.023)} \ln P^{ng}_{it} + \underbrace{0.060}_{(0.055)} \ln \mathrm{GDP}_{it} + \alpha_i + \lambda_t + v_{it}$$

HC1 SEs clustered at state level.

	Value
Short-run price elasticity	\$-\$0.195
Long-run price elasticity: $\hat{\beta}_1/(1-\hat{\rho})$	\$-\$1.104

8.3 Robustness

Hawaii exclusion — coefficients barely move (SR elasticity shifts from \$-\$0.195 to \$-\$0.207). Not being driven by Hawaii.

Temporal heterogeneity — the pre/post-2021 split you flagged as interesting:

Period	SR Elasticity	LR Elasticity	Note
Full 2012–2024	-0.195**	1.104	Preferred baseline
Pre-2021	-0.075 (n.s.)	0.238	Elasticity not activated
Post-2021	-0.255***	-0.767	$T = 3$ after lag filter

Chow test: $F(4, 542) = 3.417$, $p = 0.009$ — structural break confirmed.

For the three shifts you highlighted:

1. **Price elasticity activated** (-0.075 n.s. to -0.255***): My reading is that pre-2021 electricity prices were too low and stable to trigger adjustment behaviour. Once the post-2021 energy crisis pushed prices high enough, businesses actually responded — retrofitting, efficiency investment, maybe some relocation of load-intensive operations.

2. **Gas substitution emerged** (-0.030 n.s. to +0.121***): Pre-2021, gas and electricity weren't really competing for the same commercial end uses. The global gas spike created enough price pressure that dual-fuel users switched load to electricity. This channel was dormant before and got switched on.

3. **GDP decoupled** (0.162*** to 0.135 n.s.): Partly a data issue — with only $T = 3$ after lag filtering, cross-sectional variation in GDP was compressed by the uniform post-COVID recovery, so identification is weak. But I also think there's a real story here: remote/hybrid work reduced electricity intensity per dollar of GDP, and efficiency investments (LED, smart HVAC) mean the GDP-to-electricity relationship genuinely shifted.

9 Questions for Our Meeting

1. **Cointegration method** — is the Fisher residual test (stat = 391, $p < 0.001$) sufficient to justify estimation in levels, or do I need Johansen / ARDL bounds given the mixed I(0)/I(1) orders?
2. **Gas substitution interaction** — would it be worth adding a $\ln P^{ng} \times$ Post-2021 interaction term to formally test whether the substitution channel is regime-dependent? I didn't do this yet but it seems like a natural extension.